IFRS 17 RA Calibration: Audit Trail and Summary

Oct 28, 2026 · @Karan Arora

Summary

A cost-of-capital IFRS 17 risk adjustment (RA) was built in Excel for a synthetic term and annuity book, first with the textbook (full) method and then with a faster factor method. The full method gives an **RA of €3.90m at an implied 77% confidence level**. A single flat factor understated it by 31%; age-banded factors fixed that in-sample (+3.7%) but failed out of sample (+11% to +27% on longevity) because they had diversification built in. **Standalone age-banded factors transfer almost exactly (within ±0.3%)**, so the corrected design applies diversification each year, outside the factors. The Python build will follow that design, using the Excel workbook as its benchmark.

All numbers are illustrative: 6% cost of capital, Solvency II-style stresses, synthetic portfolio. None is any company's calibration.

Why two methods

Both methods compute the same thing: RA = 6% × Σ capital(t) × discount factor (IFRS 17 para 37, B86–B92). They differ in how the future capital path is found.

	Textbook (full) method	Factor method
How capital(t) is found	Re-run every stress (mortality, longevity, lapse, expense) at every future year	Capital = factor × exposure, with factors calibrated once
Accuracy	Exact for the model	Approximate; depends on the factors being stable
Speed	Slow: stresses × years × model points	Fast: one calibration, then exposures only
Role here	The benchmark	What a real reporting process would use

The full method is too slow for a real book, especially where capital comes from an internal model, so insurers use factors. The full method was built first so the factor method could be tested against it. The point of the exercise was to find out when the

shortcut is safe.

Audit trail

Each step, newest first, with what changed and what it showed.

Date	Step	What changed	Result	Where
28 Oct 2026	Decision	Move the build to Python, using the Excel workbook as the benchmark	Corrected design agreed (see Diagnosis)	Python (to build)
28 Oct 2026	Standalone factor test	Longevity bands calibrated on standalone capital, frozen at age 65, applied to books starting at 70 and 75	Error −0.3% (70) and −0.2% (75): standalone factors transfer	Age bands sheet, columns O–Q
28 Oct 2026	Out-of-sample test	Euler band factors frozen at age 65; annuity start age changed to 70 and 75	Total RA error +8.8% (70), +14.4% (75); longevity +10.6%, +27.2%	Age bands sheet, column N
28 Oct 2026	Age bands rebuilt by hand	Bands built step by step in own workbook	Matches the reference build: +3.7%	Own workbook

Date	Step	What changed	Result	Where
25 Oct 2026	Age-banded longevity factors	Eight 5-year attained-age bands (capital ÷ exposure per band, SUMIFS)	RA €4.05m, error +3.7%, run-off confidence 77.9%	IFRS17_RA_Cost_of_Capital_Workbook_AgeBands.xlsx
25 Oct 2026	Factor RA by group	Factor RA split into term and annuity	Term +10% (€1.35m vs €1.22m), annuity −49% (€1.36m vs €2.68m)	Factor method sheet
24 Oct 2026	Factor-method confidence sheet	Own sheet comparing the factor RA with the full-path σ	69.6% (vs 73.7% if σ used the factor path, rejected as flattering)	Confidence Factor method sheet
24 Oct 2026	Full step-by-step workbook	Per-policy BEL by backward recursion, stresses re-run each year, correlation, Euler allocation, factor method, confidence	Full RA €3.90m; flat-factor RA €2.70m (−31%); confidence 77% run-off vs 99% one-year	IFRS17_RA_Cost_of_Capital_Workbook.xlsx
24 Oct 2026	Book rebalanced	Annuity policies reduced from 2,000 to 300 so neither group dominates	Removes negative Euler shares for the term group	Inputs sheet

Date	Step	What changed	Result	Where
23 Oct 2026	First example	Simple factor sheet: Solvency II-style stresses, Euler allocation, 6% CoC	RA €6.89m, confidence about 78% (normal approximation)	IFRS17_RA_Cost_of_Capital_Example.xlsx

The Excel solution

The workbook runs from inputs to the disclosed confidence level, one sheet per step.

Sheet	What it does	Key formula or logic
Inputs	Assumptions: term book (10,000 policies, age 40, 20 years), annuity book (300 policies, age 65), Gompertz–Makeham mortality, stresses, Art. 136 correlations, 6% CoC	Blue cells are inputs
Term	Per-policy BEL on best estimate and each stressed basis; capital and exposures	BEL(t) = expense − premium + q·SA·v + (1−q)(1−lapse)·v·BEL(t+1); capital = in-force × (stressed − best-estimate BEL)
Annuity	The same for the annuity, with the longevity and expense stresses	BEL(t) = expense + (1−q)·annuity·v + (1−q)·v·BEL(t+1)
Capital	Diversified capital each year; standalone by group; Euler and proportional allocation	$C = \sqrt{(\Sigma_i C_i \times \Sigma_j \rho_{ij} C_j)}$; Euler share = $C_i \times (\Sigma_j \rho_{ij} C_j) \div C$
RA	RA by backward recursion, total and by group	RA(t) = CoC × C(t) × v + v × RA(t+1)
Factor method	One factor per risk at t = 0, then capital = factor × exposure	Factor = Euler capital at t = 0 ÷ exposure at t = 0

Sheet	What it does	Key formula or logic
Age bands	Longevity factors by 5-year attained-age band; out-of-sample and standalone tests	Band factor = SUMIFS(capital) ÷ SUMIFS(exposure) per band; INDEX/MATCH picks the band
Confidence	Implied confidence level for para 119, one-year and run-off views	σ = capital ÷ 2.576; run-off σ = $\sqrt{\Sigma(\sigma_t \cdot v^t)^2}$; confidence = NORMSDIST(RA ÷ σ)
Confidence Factor method	The same for the factor RAs, always against the full-path σ	σ measures the real risk, whatever shortcut produced the RA
Summary	Key results and checks (all checks should be 0)	BEL identity, Euler allocation and allocated RAs add up

Results

In-sample, age-banded factors come within 4% of the full method; out of sample, only standalone factors hold.

In-sample (annuity book starting at 65)

Method	RA	Error vs full	Implied confidence (run-off view)
Full (textbook) method	€3.90m	—	77.1%
Flat factor at t = 0	€2.70m	−31%	69.6%
Age-banded factors (Euler)	€4.05m	+3.7%	77.9%

The diversification benefit is 29% (standalone RAs €5.46m vs diversified €3.90m). The one-year confidence view gives 98.9%, which overstates for long-dated annuities.

Out of sample (bands frozen at age 65; longevity RA error vs full method)

Annuity start age	Euler (diversified) bands	Standalone bands
65 (in-sample)	+0.4%	+0.3%
70	+10.6%	−0.3%
75	+27.2%	−0.2%

Diagnosis and fix

The diversified factors were labelled by age, but they had also absorbed when the term book runs off.

1. **Flat factor too low.** Longevity capital per unit of PV annuity payments rises with age: 3.0% at 65, 8.5% at 80, 23% at 100. A factor fixed at 65 understates the later years.
2. **Age bands fixed that in-sample.** One exposure-weighted factor per 5-year band follows the rise with age.
3. **But the bands were built from Euler (diversified) capital.** Longevity is partly offset by the term book's mortality risk (correlation −0.25). The term book ends after 20 years, so in the age-65 book there is no term business left once annuitants reach 85. That is why the Euler and standalone band factors are identical from 85 onwards (12.8%, 15.9%, 19.6%, 24.0%), and lower before (3.5% vs 5.5% at 65–69).
4. **In an older book the timing shifts.** With annuitants starting at 75, term business is still in force at ages 85–94, so real diversification applies there, but the frozen factors assume none. The factors overstate capital, more so the older the book.
5. **Proof.** Standalone factors, with no diversification inside them, transfer within ±0.3%.

Fix: keep diversification out of the factors.

1. Calibrate standalone factors by band for each risk: attained age for mortality and longevity, policy duration for lapse.
2. Each year, standalone capital per risk = factor × exposure.
3. Apply the correlation matrix to those capitals each year.
4. Allocate to groups (Euler), then apply the cost of capital.

Assumptions and limitations

- **Synthetic, illustrative data.** One model point per group; Solvency II-style stresses stand in for a 1-in-200 capital model. Not any company's calibration.
- **Flat 3% discount rate**, and a 6% cost of capital (the Solvency II rate falls to 4.75% from 2027).
- **Expense stress is level only.** The Solvency II +1% inflation stress is not yet included.
- **Confidence level by normal approximation.** The run-off view assumes independent yearly shocks; persistent longevity risk would widen σ and lower the confidence level. A Monte Carlo simulation would replace this.
- **Term-book factors are still flat.** Mortality and lapse have not been banded or tested out of sample; the term group is about 10% off in-sample.

- **In-sample calibration.** The bands were fitted and first tested on the same book; only the longevity out-of-sample test has been run so far.

Next steps: Python build

The Python model reproduces the Excel benchmark first, then adds the corrected factor method.

- [ ] `inputs.py`: assumptions from the Inputs sheet
- [ ] `projection.py`: per-policy BEL by backward recursion on any basis, vectorised over model points
- [ ] `capital.py`: standalone capital per risk and year, diversified capital ($\sqrt{(c^T R c)}$), Euler allocation
- [ ] `ra.py`: RA recursion, total and by group
- [ ] `factors.py`: standalone factors by band (age for mortality and longevity, duration for lapse), diversification applied each year
- [ ] `confidence.py`: run-off σ and implied confidence level
- [ ] `tests/`: Excel benchmarks as fixed checks

Benchmark	Expected (Excel)
Term BEL per policy at t = 0	−1,299.68
Annuity BEL per policy at t = 0	203,497
Diversified capital at t = 0	€4.38m
Full-method RA	€3.90m
Implied confidence, run-off view	77.1%
Banded longevity RA (Euler), age 65	€2.66m
Standalone banded longevity error, ages 70 / 75	−0.3% / −0.2%

After the benchmarks pass: run the corrected method on books with different term and annuity starting ages, then on the SCR calculator's 12,000-policy book.